Markov State Modeling of the PHF6 peptide of Tau protein

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In the following work, I investigate the dynamics and conformational landscape of the PHF6 peptide and its mutation V306M using molecular dynamics simulations (MD) and Markov State Models (MSM). The objective of the project is to use a standard pipeline to retrieve information about the dynamics of the two peptides, in order to discuss the dynamics and stability of the two systems. To do so, I generated trajectories in the phase space and then used MSM to compute transition matrices and then compare the two different peptides. Despite limited resources and sampling, I've been able to show how the mutant exhibits additional slower relaxation processes and shifts in free energy basin populations, as well as the presence of new slow basins.

Keywords: Intrinsically Disordered Proteins, PHF6, V306M, molecular dynamics, Markov state model

I. INTRODUCTION

Intrinsically disordered proteins (IDP) are getting more and more attention and study, especially after the advent of very powerful chips to simulate molecular dynamics and the enrichment of databases with predicted structures [1]. The peculiarity arises from the fact that segments of these proteins are not at equilibrium and the free energy landscape showcases many shallow minima from which the system can escape and reach another one, thus changing the structure of the protein, rather than having a single fixed state [2].

TAU protein is a classic example of IDP and is implicated in neurodegenerative diseases, such as Alzheimer's or dementia. In particular its K18 segment is highly disordered and subject of intense studies [3]. Given the long sequence of K18 (over 120 residues [4]), in this project I focused on a shorter motif on the sequence, the PHF6 peptide (VQIVYK), to see what metastable configurations it presents and how they're altered with mutations [5]. I chose the V306M mutation, where the initial Valine (V) has been mutated in Methionine (M) (Fig. 1). Mutations in the MAPT gene encoding TAU, including FTDP-17–associated variants such as V306M, have been linked to the rise of dementia in patients [6, 7].

The main question I aim to investigate is how the mutation of the first amino acid changes the structure and dynamics of this peptide. In order to answer this question, MD simulations were carried out to generate trajectories for both peptides and these aritificial data have been used to estimate transition rates using MSM. This will allow to investigate the free energy landscape and find (meta)stable configurations.

In the following sections, I will describe the simulation and MSM construction protocol and analyze the resulting conformational landscapes and kinetic properties. Finally, a discussion on how the V306M mutation alters the dynamical behavior of the PHF6 peptide is presented.

II. METHODS

A. Dataset assembly

The first step was to generate the trajectories for the simulations. This was achieved first by downloading the TAU sequence from Alphafold [8, 9] and then by using the OpenMM library for Python [10]. I identified the VQIVYK motif in the residue chain, isolated it and then added the missing hydrogens using PDBFixer [10] In particular, for both peptides, I simulated the systems for 15ns in an implicit solvent at 300K with the Amber14 force field and saving configurations every 15 ps. This way I was able to obtain 90 trajectories for each peptide, evolving a common topology file using a Langevin integrator with a timestep of 2 fs. The first 400 ps were discarded as equilibration time. Two resources were used for computations: a NVIDIA-SMI 555.51 GPU using CUDA and a 16x4GB core virtual machine on CloudVeneto [11].

As will be discussed in section IV, I then simulated more trajectories for the V306M mutation, starting from the 4 states with highest probability and extending the simulation time to 100ns.

B. MSM construction

First, the data was preprocessed, embedding dihedral angles as sine and cosine and computing pairwise Cα distances. Then, using the Deeptime package [12], a TICA analysis was computed using the same model for both peptides, in order to effectively capture how the markovian dynamics changed due to the mutation and reduced on the three slowest components. Finally, the fitted trajectories were clustered in 40 groups using KMeans.

After the dimensionality was reduced and the trajectories discretized, it was possible to run the MSM on the data, again using the Deeptime package [12]. I chose a lag time τ of 40 frames (600 ps) to capture the markovian nature of the dynamics; The output of the MSM used were the transition matrix T and the stationary distribution π such that $\pi T = \pi$. From the transition matrix, it was then possible to infer the timescales of the process

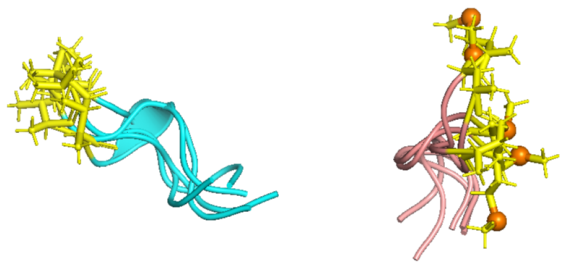


FIG. 1: *Representative ensembles of the 5 most populated MSM microstates for PHF6 wild type (left) and V306M mutant (right), generated with PyMol [14]. Structures are aligned within each ensemble and displayed as cartoons. The mutation site (residue 1) is highlighted in yellow; sulfur atoms in methionine are shown as orange spheres. Compared to the wild type peptide, the V306M mutation in these simulations appears to have a more rigid and closed structure, consistent with increased metastability and slowed dynamics observed in the MSM analysis.*

as

$$t_i = -\frac{\tau}{\ln|\lambda_i(\tau)|} \tag{1}$$

with eigenvalues λ_i giving the slowest relaxation timescales. It was then also possible to compute the free energy $F = -k_B T ln(\pi)$ [13].

C. Statistical analysis

As a measure of good fitting of the MSM, I focused primarily on the convergence of timescales as a function of the lag bootstrapping over trajectories to smoothen out rare transitions. Finally, a Chapman-Kolmogorov test (CK) was performed selecting 4 macrostates as an aggregate over microstates using Perron-cluster cluster analysis (PCCA) and comparing the credibility of the transition matrix T with an empirical one, built as normalized counts of transitions from trajectories [13]. Agreement between predicted and empirical macrostate dynamics was quantified using the L1 norm between transition probabilities evaluated at multiples of the MSM lag time. Summary statistics included the median and 90th-percentile L1 deviations across lag multiples.

III. RESULTS

A. Low-dimensional embedding reveals distinct conformational regions

Projecting the trajectories onto the first two time-lagged independent components reveals different conformational organization for the two peptides (Fig. 2a).

TABLE I: MSM diagnostic summary at lag $\tau = 40$ frames. CK metrics computed on 4 macrostates. 1 frame equals to 15 ps.

Metric	PHF6 WT	PHF6 V306M
Active microstates	35	40
Slowest timescale t_1 (frames)	162	2248
CK credibility	0.593	0.938
CK median L1	0.665	0.049
CK p90 (L1)	0.738	0.106

The PHF6 wild type, evolves mainly on fast timescales, with very rare transitions on a long timescale over the second principal component, but overall its dynamics is uniform over a short region of fast timescales. Changing the first residue of the peptide, slows down the dynamics of the peptide. It still evolves on the timescales associated to the wild type peptide, but in addition it has new dynamics, which are slower on both principal components.

These observations suggest that the mutation reshapes the slow collective modes of the system, favoring kinetically isolated conformational states, as will be discussed in the next paragraph.

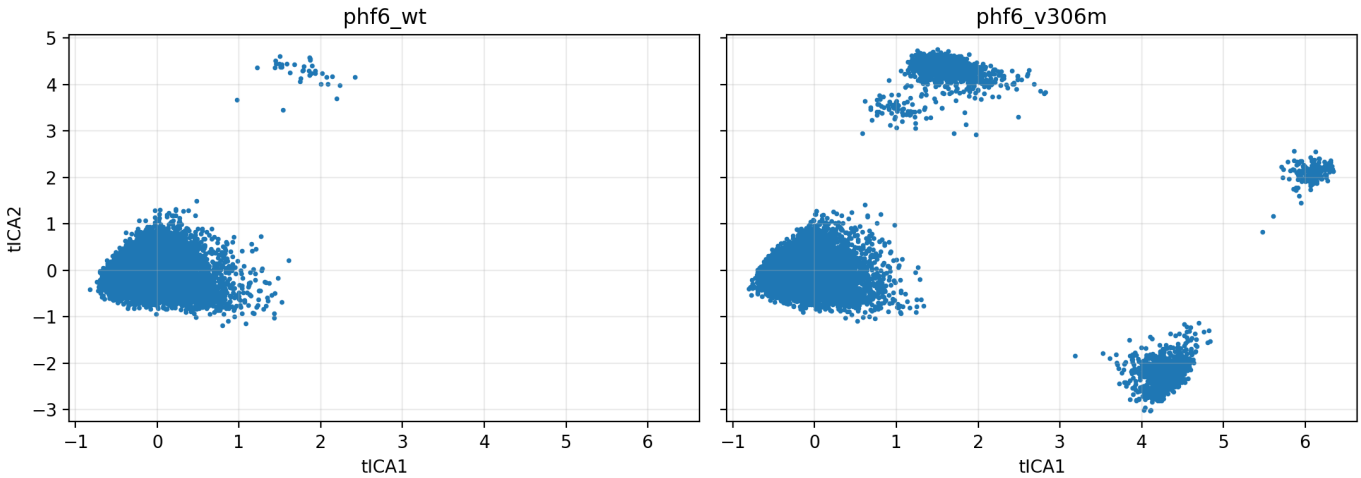
B. MSM reveals mutation-dependent changes in metastable basins

Free-energy surfaces reconstructed from MSM stationary distributions further highlight the impact of the V306M mutation (Fig. 2b). While the wild-type peptide populates a (large) but connected basin, the mutant system occupies a smaller area on fast timescales and shows different free energy minima on slower timescales. It is worth noticing how even the fast basin in common with the wild type, is shifted toward slower timescales on the second component.

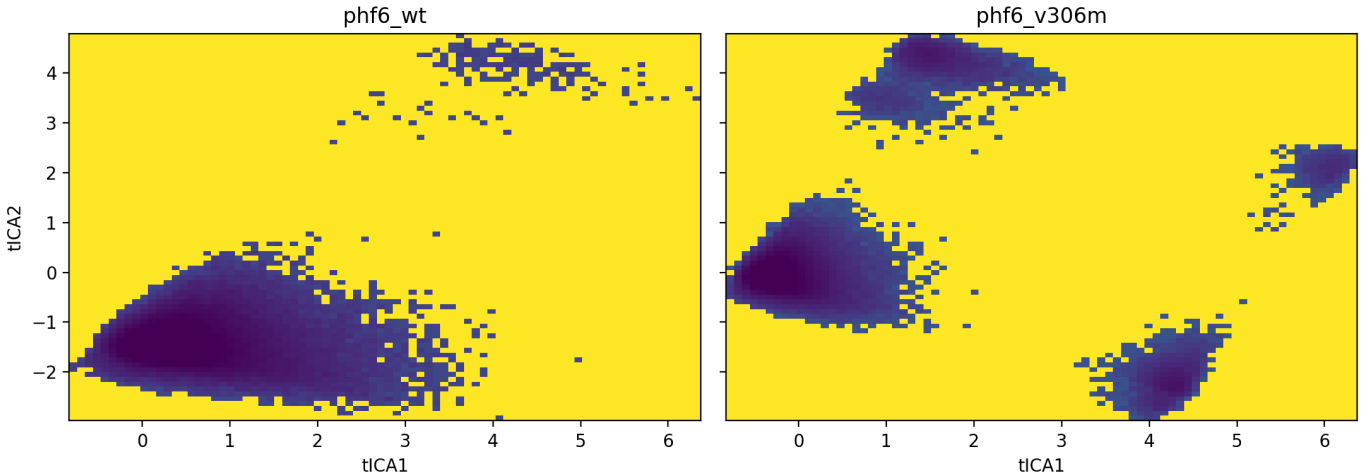
This redistribution of thermodynamic weight suggests that V306M stabilizes specific backbone arrangements at the expense of conformational diversity, consistent with a loss of dynamic flexibility.

C. Markovian dynamics of the peptides and implied timescales

Looking at table I, it is possible to infer some information on the MSM. In particular, even with a reduced cluster size of 40 centers, the wild type peptide shows transitions only between 35 of them, thus suggesting a limited range of dynamics. Even though the V306M mutation has a broader dynamics, its timescales are so slow that my simulations weren't able to capture them. Thus, the mutation slows down the dynamics by more than an order of magnitude (from $\sim$ 160 frames to more than



(a) *Projection onto the two TICA principal components for PHF6 conformations (wild type on the left, V306M on the right). It is evident how the mutation changes the system to evolve on slower dynamics, on both the principal components.*



(b) *Discrete state free energy derived from MSM stationary populations, projected after TICA analysis, for wild type on the left, V306M on the right. Lower regions correspond to more populated metastable conformations (colored in dark blue). The energy scale is the same for both peptides (not reported here as it is in a.u.). It is again clear, how the wild type is mostly confined in a small region, while the mutation explores a larger and slower dynamics, identifiable in these different low-energy basins.*

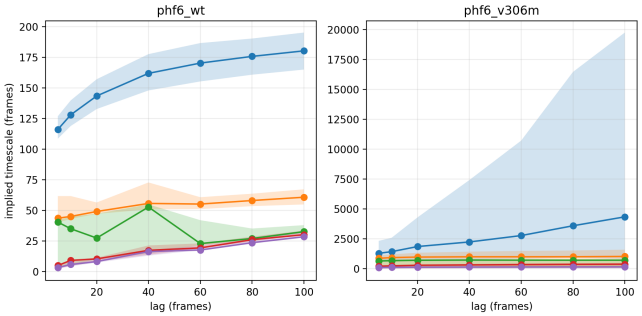


FIG. 3: *Plot of the implied time scale (ITS) as a function of lag τ for the wild type peptide (left) and the V306M mutation (right). Shaded regions indicate uncertainties, computed with bootstrap. The colors represents the different timescales, going from the slowest to the fastest the color coding is: blue, orange, green, red, purple.*

2000), implying that a mutated peptide has a strong impact on the dynamics.

Finally, it is possible to make some remarks on the markovian nature of these systems on a 40 frame lag scale. The credibility of the CK test for the two peptides, indicates that the wild type one has a faster dynamics, which wasn't fully captured by the simulations, while after the mutation, this dynamics is slowed down so much that a markovian approximation fits the data well. It is still worth noticing how the CK test indicates a higher credibility for V306M, but the area sampled is relatively small, as we don't sample over rare transitions, thus showing consistency, but only in the small sampled region.

IV. DISCUSSION AND CONCLUSION

In this work, I used MSMs to compare the conformational dynamics of the PHF6 peptide in its wild-type form and with the V306M mutation.

After the analysis, it is possible to draw some insights on the differences between the two peptides, though the limitation of resources hindered the collection of longer trajectories, in particular for the V306M mutation. The implied timescale (ITS) test (Fig. 3) reveals a qualitative difference between PHF6 WT and V306M. For the WT peptide, the slowest implied timescales increase moderately with lag time and begin to approach a plateau, indicating that a Markovian description becomes approximately consistent beyond a few hundred picoseconds. In contrast, V306M exhibits substantially larger slow timescales and a pronounced lag-time dependence of the slowest mode, together with very wide uncertainty at larger lags, consistent with rare transitions and undersampling of the slow processes. This behavior indicates insufficient statistical support for the slowest eigenmodes, rather than a failure of the MSM formalism itself, which is instead consistent for the wild type peptide [15, 16].

As previously stated, after collecting these results, I ran longer simulations (100 ns) for the mutated peptide, starting from the 4 seeds that had the highest transition counts. Though these new trajectories helped populating the free energy basins, they didn't improve the analysis of ITS, which were still well out of range of my simulations. Importantly, the observed limitations arise from insufficient sampling of slow processes, rather than from a breakdown of the MSM framework itself.

Nonetheless, it is possible to argue that the effects of the mutations are visible and relevant even on short timescales and a relatively large sample lag, because new isolated basins arise in the free energy landscape, each of them populated with a quite large number of counts. This is consistent with previous experimental studies showing that FTDP-17–associated tau mutations alter tau biochemical and conformational properties [6].

As a remark for future works, it is worth noticing that larger and faster resources could easily retrieve these slow features of the mutated peptide, as well as completely capture the fast dynamics of the wild type by saving trajectories every few picoseconds. It would then be interesting to enlarge the amino acids section around the PHF6 peptide, until eventually capturing the whole K18 chain dynamics.

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